University of Central Punjab
Faculty of Information Technology

BSCS

PROGRAM (S) TO BE EVALUATED

A. Course Description

Course Code	TBD
Course Title	Cyber Security
Credit Hours	3
Prerequisites by Course(s) and Topics	NA
Post requisites by Course (s) and Topics	
Assessment Instruments with Weights	Assignment (15%)—At least 4 Quiz (15%)---At least 4 Mid Term (20%) Class Participation(10%) Final Term (40%)
Semester	Spring 2026
Course Instructor	Muhammad Bilal Khan
Course Instructor Email	Muhammad.bilalkhan@ucp.edu.pk
Course Coordinator	Muhammad Bilal Khan
Textbook	1. Computer Security: Principles and Practice, 4th Edition, William Stallings and Lawrie Brown
Reference Material	1. Computer Security Fundamentals by Chuck Easttom, 4th edition or latest 2. Security+ Guide to Network Security Fundamentals, by Mark Ciampa, 5th Edition *3.* Security in Computing by C.P. Pfleeger, Prentice-Hall, 4th Edition or Latest
Course Introduction & Objectives	This course offers an introduction to key cybersecurity concepts, focusing on common threats, vulnerabilities, and risks across web applications, networks, software, and mobile platforms. Students will learn essential terminology and concepts in the cybersecurity field, while also gaining the ability to identify various types of malwares and understand their impact on computers and networks. This course provides a strong foundation for understanding

	cybersecurity fundamentals.
Course Objectives	

CLO No.	Course learning Outcomes	Blooms Taxonomy Level	PLOs
CLO-1	**To be able to identify computer system threats**	1	PLO 1- Academic Education
CLO-2	**To be able to identify Malware attacks, and understand the stages of attack and payloads**	2	PLO 1- Academic Education
CLO-3	**Implement various cryptographic techniques and simulate attack scenarios**	3	PLO 1- Academic Education

Bloom Taxonomy: 1: Remember, 2: Understand, 3: Apply, 4: Analyze, 5: Evaluate, 6: Create

Week#	Lecture#	Topic	Instruments	
1	1	- Overview of cybersecurity - Importance of cybersecurity		
	2	- Basic terminology - History of cyber threats		
2	3	- Introduction to computer system threats - Types of threats (internal vs external) - Attack vectors	Quiz 1	CLO-1
	4	- Common cyber threats - Service attacks	Assignment 1	CLO-1
3	5	- Definition of vulnerabilities - Common system vulnerabilities - Introduction to CVE		
	6	Vulnerabilities in systems	Quiz 2	CLO-2
4	7	- Types of malwares - Malware infection mechanisms.		
	8	- Impact of malware on systems - System errors		
5	9	Introduction to Malware Attack Lifecycle Stages of a Malware Attack		
	10	- Payloads - Defending Against Malware	Assignment 2	CLO-2
6	11	- Network Defenses - Firewalls - Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS)		

	12	- Benefits and Limitations - Best Practices for Network Defenses	Quiz 3	CLO-1, CLO-2
7	13	- Introduction to cryptography - Symmetric vs Asymmetric		
	14	- Encryption - Public and private keys		
8	15	- Encryption algorithms (AES, DES) - Hashing algorithms (SHA, MD5)		
	16	Digital signatures		
9		**Mid Term**		CLO-1, CLO-2
10	19	- Introduction to network security Firewalls		
	20	- Virtual Private Networks (VPNs Network segmentation		
11	21	Common web application vulnerabilities (SQL injection, XSS, CSRF)		
	22	- Secure coding		
12	23	- Penetration testing methodology - Reconnaissance	Assignment 3	CLO-2, CLO-3
	24	- Enumeration techniques. - Scanning	Quiz 4	CLO-3
13	25	- Exploiting vulnerabilities - Creating custom payloads		
	26	Post-exploitation activities.		
14	27	- Incident response lifecycle - Detection - Containment	Assignment 4	CLO-1 CLO-2 CLO-3
	28	- Eradication - Recovery - Reporting		
15	29	- Cloud Security Challenges and Solutions - Securing Cloud Environments (IaaS, PaaS, SaaS)		
	30	- Cloud Compliance and Best Practices (Data encryption, access controls, incident response in the cloud)	Quiz 5	CLO-2, CLO-3
16	31	- Cyber Laws and Regulations - Ethical, Legal, and Professional Issues in		
	32	- Cybersecurity - Ethical Hacking and Ethical Considerations - Professional Certifications in Cybersecurity		

	Final Term		CLO-1, CLO-2, CLO-3